45. Evaluate $\int_0^1 \int_0^1 x^2\, y dx dy$

46. Evaluate $\int_1^2 \int_2^4 6x\, y^2\, dx dy$

47. Evaluate $\int_0^2 \int_0^2 4mnxy dx dy$

48. $\int_1^2 \int_1^3 (4x-2y)\, dx dy$

ANSWERS:

I. 1. (a) , 2. (c) , 3. (c) , 4. (c) , 5. (a) , 6. (c) , 7. (b) , 8. (b) , 9. (a) ,
10. (b) , 11. (a)

II. 12. n! 13. 24 , 14. 45, 15.1 , 16. 1330 , 17. $\frac{13}{10}$, 18. e^x ,
19. $\log x$, 20. x^6+c, 21. $\frac{1}{11}$

III. 22. 60, 23. (i) 24 , (ii) 5040 , (iii) 9900 , 24.120 , 25. 90,
26. (i) 210 , (ii) 1540 , (iii) 4950 , 27. (i) $\frac{3}{10}$, (ii) 10 , 28. $e^x\ (x^2+2x)$

IV. 29. $\frac{11}{2!2!}$, 30. 350 , 31. 20 , 32. 30 , 33. (i) $e^x\ (x^2+2x-5)$,
(ii) $\frac{x^2+14x+1}{(x+7)^2}$, 34. $\frac{3}{4}x^4-\frac{2}{3}x^3+3x^2-7x+c$ 35. $2\log x - \frac{2}{x} - e^x + 3x + c$
36. $\frac{3}{2}$, 37. 0, 38. 7

V. 39. 497574 , 40. (i) $4(13C_3.39C_1)$, (ii) $(13C_1)^4$, (iii) $26C_2.26C_2$, 41. 48,
42. (i) $e^x\ (x^3-x^2-6x+2)+5(3x^2-8x+2)$, (ii) $\frac{7x^2-16x-56}{(x^2+8)^2}$
43. (i) $\frac{x^2}{2}-5x+6\log x+c$, (ii) $\frac{4}{3}x^{\frac{3}{2}}-\frac{3}{x}+c$, 44. (i) 1, (ii) 12 log2
45. $\frac{1}{6}$, 46. 84 , 47. 16 mn , 48. 24